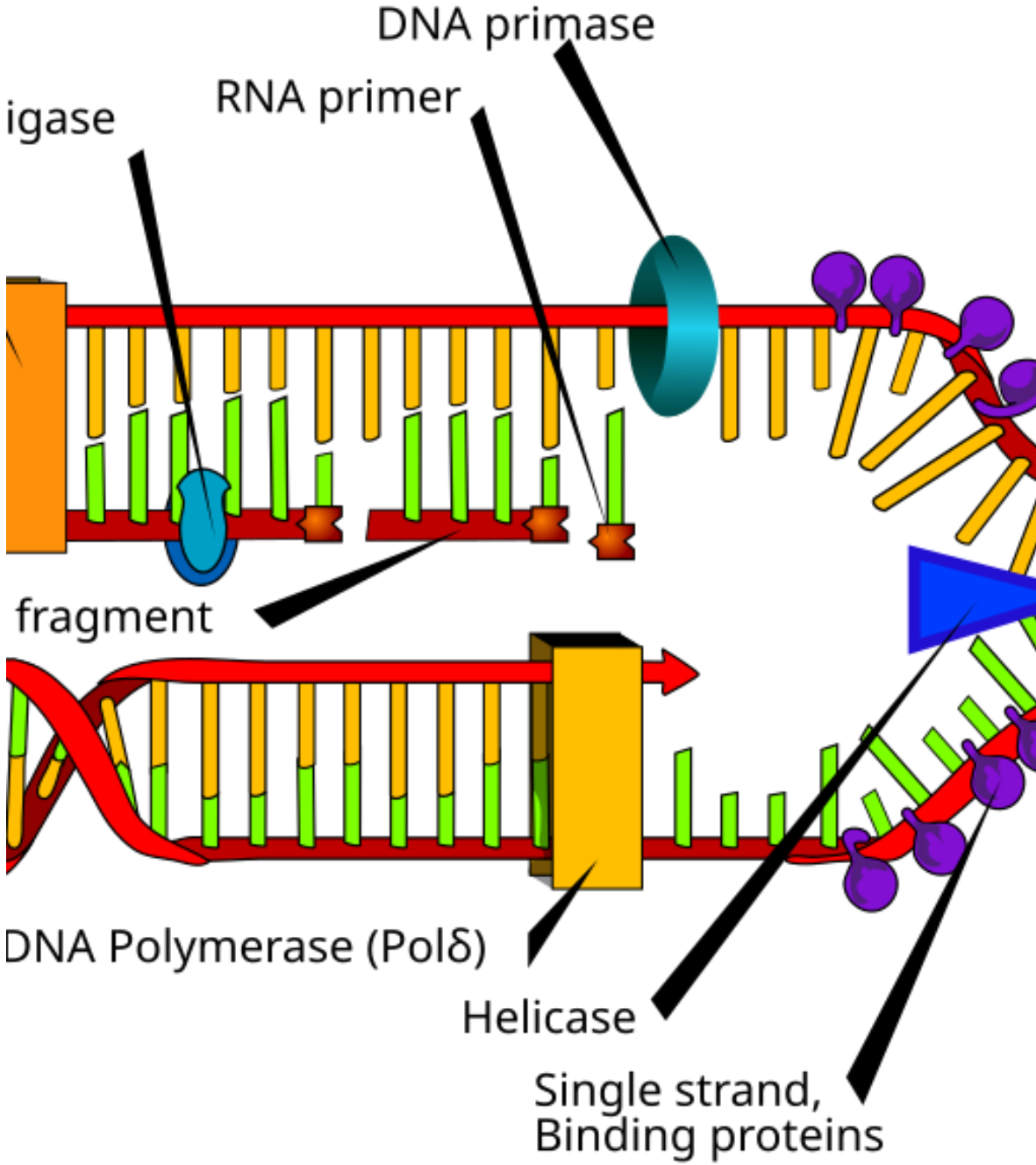
DP Biology – D1.1 (HL extension)

# Inside DNA replication

- How do cells make new DNA?
- What biotechnology has understanding DNA replication made possible?



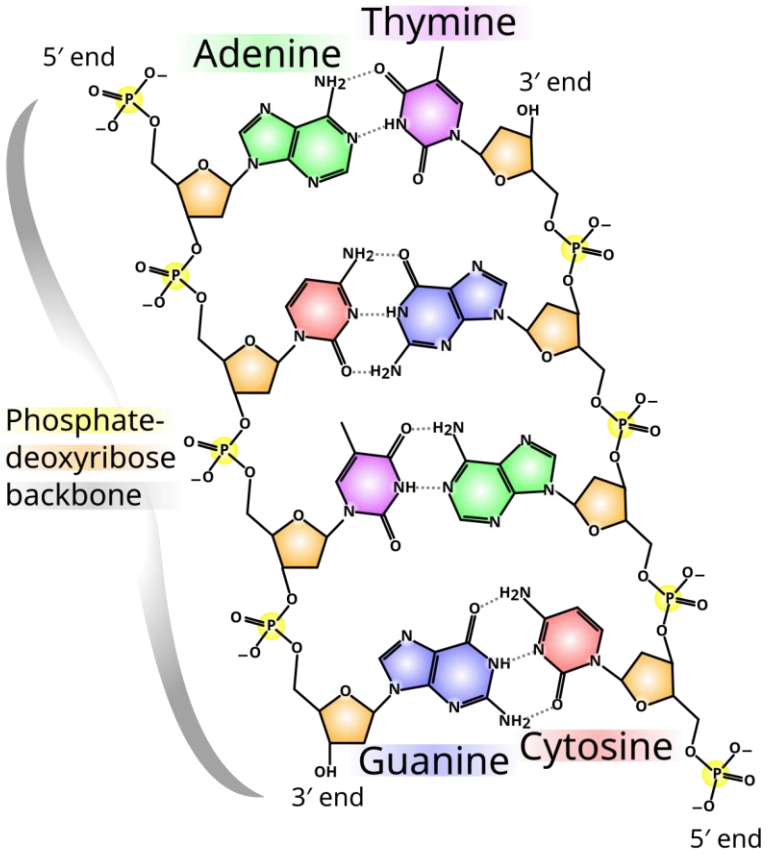

HL statements D1.1.6–9. Textbook pp. 604–614.
Image: DNA replication. Mariana Ruiz Villarreal, public domain.

# Objectives for each class

| Class | Statements | By the end of the class, you can... |
|---|---|---|
| **1** | D1.1.6 | Explain why DNA polymerases can only build DNA in the 5′ to 3′ direction |
| **2** | D1.1.7–8 | Compare the leading and lagging strands, and explain the roles of primase, DNA polymerases I and III, and ligase |
| **3** | D1.1.9 | Explain how DNA polymerase III proofreads, and why this matters |

# DNA polymerase works in one direction

D1.1.6




DNA strands are antiparallel. CC BY-SA 3.0.

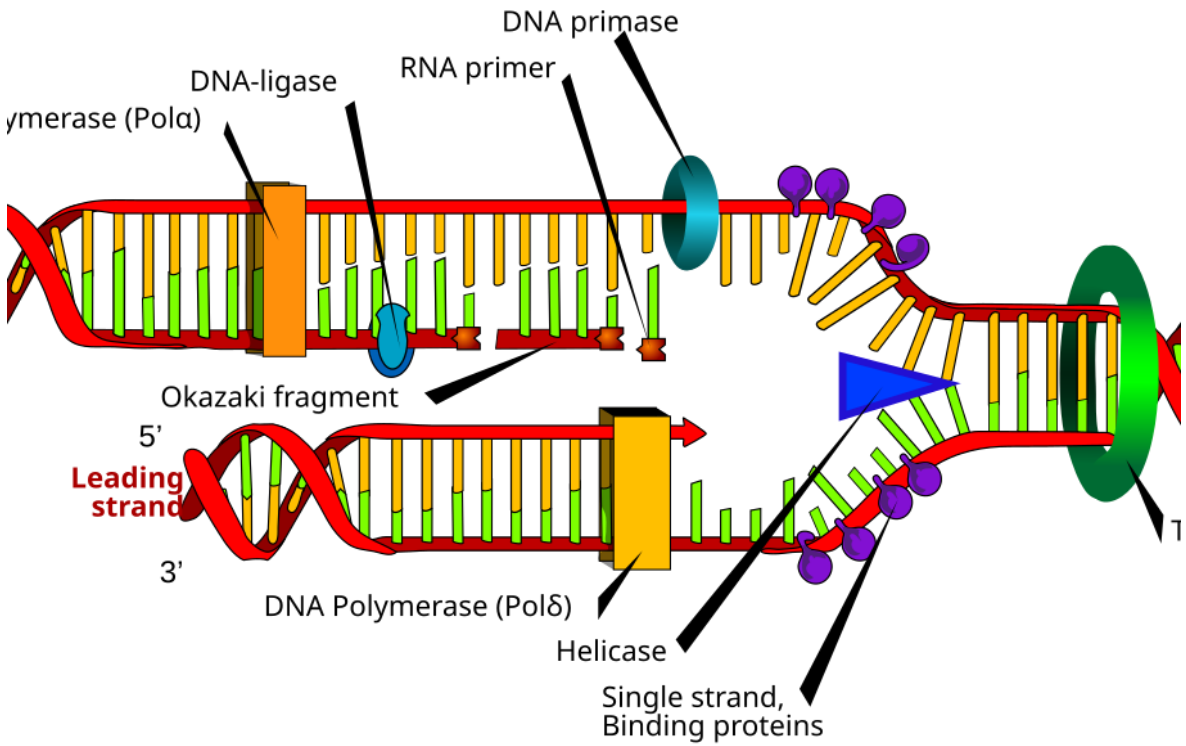
- A free nucleotide arrives as a nucleoside triphosphate
- DNA polymerase links its 5′ phosphate to the free 3′ -OH at the end of the growing strand
- Two phosphates are released, providing the energy
- So the new strand always grows **5′ to 3′**, and the template is read 3′ to 5′
- DNA polymerase cannot start a new strand: it needs an existing 3′ end

### Why only one way?

The energy-carrying phosphates are on the incoming nucleotide's 5′ end. Adding to the 5′ end of the strand would need the energy to be on the strand instead, which would be lost every time a wrong base was removed.

# Leading and lagging strands

D1.1.7




| | Leading | Lagging |
|---|---|---|
| **Synthesis** | Continuous | Discontinuous |
| **Direction** | Towards the fork | Away from the fork |
| **Primers** | One | Many, one per fragment |
| **Pieces** | One long strand | Okazaki fragments |

Replication fork, with eukaryotic enzyme names (Pol). Public domain.

## Why?

Both new strands grow 5′ to 3′. The strands are antiparallel, so as the fork opens one new strand can follow it, while the other must be made backwards in short pieces.

# The enzymes of replication

D1.1.8

| Enzyme | Role (prokaryotes) |
|---|---|
| **Helicase** | Unwinds the helix and separates the strands by breaking hydrogen bonds |
| **DNA primase** | Makes a short RNA primer, giving a 3′ end to build on |
| **DNA polymerase III** | Adds DNA nucleotides to the primer, 5′ to 3′, by complementary base pairing |
| **DNA polymerase I** | Removes the RNA primers and replaces them with DNA |
| **DNA ligase** | Joins the sugar-phosphate backbones of adjacent fragments |

### Order on the lagging strand

Primase, then polymerase III, then polymerase I, then ligase.

# Build a replication fork

D1.1.7–8

1. In pairs, draw a replication fork with both template strands labelled 5′ and 3′. Add each enzyme where it acts.
2. **Predict** what would happen on each strand if a bacterium had no working primase.
3. **Predict** what would be found in a cell with no working ligase.
4. **Explain** why the lagging strand needs many primers but the leading strand only one.

**Hint**

Without ligase, look for many short pieces. Without primase, look at what DNA polymerase needs to begin.

# Proofreading by DNA polymerase III

D1.1.9

A wrong nucleotide is added at the 3′ end → The mismatch distorts the helix → **Polymerase III moves back and removes the wrong nucleotide** → It adds the correct one and carries on

| Stage | Errors per base, roughly |
|---|---|
| **Base pairing alone** | 1 in 100 000 |
| **After proofreading** | 1 in 10 million |
| **After later repair** | 1 in a billion |

### Why it matters

A bacterial genome has about 5 million bases, a human cell 6 billion. Without proofreading, every replication would add many mutations. Taq polymerase, used in PCR, does not proofread, so PCR copies can contain errors.

# Prokaryotes and eukaryotes

D1.1.6–9

| | Prokaryotes | Eukaryotes |
|---|---|---|
| **Origins of replication** | One per circular chromosome | Many per chromosome |
| **Replication forks** | Two, moving in opposite directions | Many |
| **Enzymes** | Polymerases I and III, primase, ligase | More polymerases, but the same principles |

### Why many origins?

A human chromosome is far longer than a bacterial chromosome. With only one origin, copying it would take weeks; many origins let the whole genome be copied in a few hours.

# Check-in: replication at HL

D1.1.6–9

1. Explain why DNA polymerase can only build a strand in the 5′ to 3′ direction.
2. Distinguish between the leading and lagging strands.
3. State the roles of primase, DNA polymerase I and DNA ligase.
4. Explain how DNA polymerase III proofreads.
5. Explain why Okazaki fragments are needed.

**How to use this**

Answer without your notes, then compare with a partner.

# Working with data: proofreading and errors

D1.1.9

Approximate error rates in DNA replication (published values, rounded).

| Stage | Errors per base copied |
|---|---|
| **Base pairing alone** | 1 in 100 000 |
| **After proofreading by DNA polymerase III** | 1 in 10 million |
| **After mismatch repair** | 1 in 1 billion |

1. Calculate how many times proofreading improves accuracy.
2. The human genome has 3 billion base pairs. Estimate the errors per replication after all checks.
3. Suggest why a low but non-zero error rate matters for evolution.

# Common mistakes to avoid

| Not this | But this |
| --- | --- |
| **DNA polymerase III adds nucleotides to the 5′ end.** | It adds them to the free 3′ end. |
| **Primers are made of DNA.** | Primers are RNA, laid down by primase. |
| **Ligase makes the new strand.** | Ligase only joins Okazaki fragments. |
| **The leading strand is made in fragments.** | The lagging strand is made in fragments. |

# Check that you can...

1. **Explain** the directionality of DNA polymerases
2. **Distinguish** between replication of the leading and lagging strands
3. **Outline** the roles of helicase, primase, DNA polymerases I and III, and ligase
4. **Explain** proofreading by DNA polymerase III

# Exit activity: 5 minutes

**On your own and without notes. Hand this in as you leave.**

1. State the direction in which new DNA is synthesized.
2. Explain why the lagging strand is made as Okazaki fragments.
3. Outline the role of DNA polymerase I.

# Image credits

DNA replication en.svg – LadyofHats Mariana Ruiz – Public domain

DNA chemical structure.svg – Madprime – CC BY-SA 3.0

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